# Infosys - Coding 11

In a special mathematical world called Dimensionia, every point in an N-dimensional space represents a possible settlement. Two rulers, Queen Petra and King Quinton, have their capitals at two different locations:

• Queen Petra's capital is at coordinates $p = (p_1, p_2, \ldots, p_n)$

• King Quinton's capital is at coordinates $q = (q_1, q_2, \ldots, q_n)$

Each ruler controls settlements based on Manhattan distance. A settlement $r = (r_1, r_2, \ldots, r_n)$ is under their influence if the sum of absolute differences is within a distance D from their capital:

• Queen's range: $d(p, r) \leq D$

• King's range: $d(q, r) \leq D$

Find the number of settlements that fall within both rulers' influence. Since the answer can be large, return it modulo 998244353.

Note: The Manhattan Distance between two points p and r in N-dimensional space is given by: $d(p, r) = |p1 - r1| + |p2 - r2| + \ldots + |pN - rN|$

**Input Format**

The first line contains an integer, N, denoting the number of dimensions.

The next line contains an integer, D, denoting the maximum Manhattan distance allowed.

Each line i of the N subsequent lines (where $1 \leq i \leq N$) contains an integer describing p[i].

Each line i of the N subsequent lines (where $1 \leq i \leq N$) contains an integer describing q[i].

**Constraints**

1 <= N <= 100

1 <= D <= 1000

-100 <= p[i] <= 100 (shown as "-100 <= p[i] <= -100", a typo for the upper bound)

**Output Format**

The output contains an integer value

**Sample Input 0**

```
1
1
1
2
```

**Sample Output 0**

```
2
```

**Explanation 0**

N=1, D=1, p=[1], q=[2]

There are 2 valid ways to construct the sequence r:

r = [1]: Differences: |1 − 1| = 0, |2 − 1| = 1. Sum of differences: 0 + 1 = 1 (valid).

r = [2]: Differences: |1 − 2| = 1, |2 − 2| = 0. Sum of differences: 1 + 0 = 1 (valid).

**Sample Input 1**

```
2
1
1
2
2
1
```

**Sample Output 1**

```
2
```

**Explanation 1**

N=2, D=1, p=[1 2], q=[2 1]

There are 2 valid ways to construct the sequence r:

r = [1, 1]: Differences: |1 − 1| = 0, |2 − 1| = 1 (for t = 1). |2 − 1| = 1, |1 − 1| = 0 (for t = 2). Sum of differences: 0 + 1 + 1 + 0 = 2 (invalid, but the DP logic ensures only valid states are counted).

r = [2, 2]: Differences: |1 − 2| = 1, |2 − 2| = 0 (for t = 1). |2 − 2| = 0, |1 − 2| = 1 (for t = 2).